

LocalStack:

Local AWS Development Made Easy

LocalStack empowers developers to harness the full power of AWS services locally, cutting costs significantly. Let's look at its benefits and find out how to build a simple AWS integration entirely on your local machine.



LocalStack is a cloud development platform that enables an AWS emulator, which runs as a Docker image on your local developer machine or your continuous integration environment. It exposes various AWS services that let you run your cloud and serverless applications without connecting to an AWS account. With LocalStack, you can access all the necessary AWS resources for your application locally and conduct automated tests in an AWS environment without the expenses of AWS developer accounts, time-consuming redeployments, or potential errors caused by remote connections.

LocalStack started as an open source project at Atlassian as an easy-to-use test/mocking framework that exposes various AWS services, such as S3, Lambda, SQS, Elasticsearch, SES, and more. The project spun off as an individual organisation and has seen two major releases — LocalStack 1.0 and LocalStack 2.0. The project now enables 80+ AWS services, encompassing various sophisticated services like Glue, Athena, RDS, QLDB, and more, to help developers improve their experience of building and testing their AWS applications.

Developing and testing cloud apps today

In the GitLab DevSecOps 2021 Survey, many developers pointed out that the lack of testing has led to delays in releasing quality software. Developers want to close the testing gap, find more errors and bugs during the development phase itself, and further enable cloud native, cross-platform development. Modern software development encompasses various cloud native dependencies — databases, virtual machines, message queues, and more. With public cloud service providers, the development and testing loop is extremely slow and tedious. Every local change must be packaged and uploaded to the cloud for testing.

The problem does not stop at just local development and testing — the closed source nature of public providers has further aggravated the workflow. Developers cannot attach a debugger to debug their code from the IDE directly; enabling debug logging is another pain, and figuring out red builds on the staging or the CI pipeline blocks other developers from testing their applications in the same time frame.

In an interesting comparison, a Hackernews user named ‘gigawatts’ equates the cloud to time share mainframes of the past, highlighting similarities such as the lack of local workstations, closed source nature, limited debugging capabilities, and delayed log-based feedback. It becomes extremely hard for managers as dev environments in the cloud are hard to manage since there is an explosion of different environments required for testing. On top of this, there is a noticeable spike in cloud costs as the resources are not cleaned properly after testing, leading to bloated cloud waste.

Getting started with LocalStack

Today cloud development is slow, tedious, and costly. LocalStack’s fully functional local cloud stack enables teams to adopt extremely efficient dev and test loops. There are plenty of advantages to using LocalStack:

- With LocalStack, you gain greater control over your environment, ensuring consistent and reliable results across different setups.